# Thesis -> Paper

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## Abstract

This thesis is being transformed into a paper

# 1 Introduction

## 1.1 Motivation and Research Context

The mean-variance framework introduced by Markowitz [1952] has shaped capital allocation for over seven decades, yet its core assumptions of stable correlations and reliable parameter estimates break down in exactly the conditions where diversification is meant to help. During market declines, correlations among S&P 500 constituents rise well above their upside levels, an asymmetry that erodes the diversification the framework is built to deliver precisely when protection matters most [Ang and Chen 2002].

# 2 Research Question

To what extent do classical and machine learning portfolio construction approaches deliver risk-adjusted outperformance and drawdown resilience across crisis episodes, and can regime detection improve them?

Sub questions:
1. How do MVO, HRP, XGB, RF and LSTM perform against market-cap weighted benchmarks, over the full sample and within crisis episodes?
2. Are the results robust to monthly, quarterly and yearly rebalancing?
3. How accurately and how early do changepoint, HMM and Wasserstein detectors identify crisis episodes?
4. Does moving to T-bills in detected crises improve each approach, and is any gain due to timing rather than lower average equity exposure?
5. Does a regime-switched combination (ML in calm, HRP in crisis) outperform each approach alone and a static mix of both?

# 3 Methodology

## 3.1 Time Frame

- 28 years, January 1998 untill December 2025

## 3.2 Rebalancing

- Rebalancing Schedule 3x, monthly, quarterly, yearly - if non trading day, next day in the month/year

## 3.3 Weight Constraints

- 1% min and 10% maximum allocation per stock per rebalancing date. - Intra rebalancing drift can exceed 10%

## 3.4 Lookback Windows

- 60 month calculation window for HRP and MVO - Fixed 60 month lookback window for all Machine Learning models | 24 month validation | + 1, 3, 12 month embargo

## 3.5 Coverage and Missing Data

- price data of min. 50% given, otherwise excluded - 5 days forward filled, otherwise missing values

## 3.6 Transaction Cost

- 0bps, 10bps, 20bps calculated. 10bps standard transaction cost - Turnover is getting calculated and reported

## 3.7 Point-in-Time

- Everything is point in time only

## 3.8 Features

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# 4 Regime Implementation

## 4.1 Changepoint Detection

### 4.1.1 Step 1

- daily VIX log changes - k-means clustering and mood rank test. is the spread of teh data after k different from before k? -> keeps the largest test statistic - if that maximum exceeds a threshold, it raises an alarm on day t (detection date). the k with the largest statistic becomes teh estimated break date t, search restarts from there - ARL0 = 10000 - Mood test uses ranks, it assumes no distribution and it reacts to changes in volatility

### 4.1.2 Step 2

- estimate teh volatility once - each detection day t, an EWMA ($\lambda$ = 0.95) of squared S&P 500 returns from t det to t gives the new regime's annualised volatility o. Always SPY returns, because 20% annual volatility is from there - Freeze it. o stays constant until the new detection. That is why your chart has steps and late exits: calm returns only after a new break is detected and the new segment's vol is below 20%. - Map it to a score: $p_{\text{crisis}} = \Phi((\sigma - 0.20)/0.05)$, where $\Phi$ is the standard normal CDF. So o = 15% gives 0.16, 20% gives 0.50, 25% gives 0.84 and etc. - shift by one day. the value on day d uses data only through d-1, which mathces the paper's one day implementation delay

## 4.2 Hidden Markov Model

### 4.2.1 Step 1

- daily log return of S&P 500 - states follow a markov chain. false L holds the probabilities of staying in each state - never the state, only returns. the model infers the state from how well each day's return fits each state's distribution - crisis is defined as teh state with teh hgher o.

### 4.2.2 Step 2

- on the first 260 days, the parameters mu, o, false L are estimated by maximum likelihood

### 4.2.3 Step 3

- update parameters daily - Each new return moves the parameters one small step toward what fits recent data best. - Older days are down-weighted by $\lambda = 1 - 1/260$, which gives about a one-year memory. - So $\sigma_{\text{calm}}$ and $\sigma_{\text{crisis}}$ drift over time. "Crisis" means high volatility relative to the last year or so, not above a fixed 20%.

### 4.2.4 Step 4

- $\alpha_t \propto \alpha_{t-1} \cdot \Gamma \cdot P(r_t)$. This is Bayes' rule: - Prior: yesterday's state probabilities, pushed through $\Gamma$ (regimes tend to persist). - Times the likelihood: how probable today's return is under each state's normal density. - Example: a −3% day is about 4.3 standard deviations under calm but only 1.9 under crisis. That makes it roughly 900 times more likely under crisis, so the probability jumps toward crisis. One quiet day after that barely moves it back, because $\Gamma$ says crises persist.

#### 4.2.5 Step 5

- $P(S_{t+1}) = \alpha_t \cdot \Gamma$. forecast = $p_{\text{crisis}}$ - It is made at the close of d−1 and stored on day d, the same one-day delay as the changepoint files. - MVO and HRP switch to min-variance when it exceeds 0.5. ML uses the value directly.

### 4.3 Wasserstein K-Means Clustering

Wasserstein k-means compares whole monthly return distributions. It sorts past months into two groups by shape, then asks which group the latest month looks more like. $p_{\text{crisis}}$ is a relative distance to the two group centres, so it is a score, not a model probability.

#### 4.3.1 Step 1

- turn each month into a point clodu - a window of 21 daily returns becomes a cloud of 21 points. In the main run each point is (S&P log return, VIX log change) - each series is divided by its expanding standard deviation up to the day before the window starts. Without this, VIX changes about 4x as volatile would dominate. Only past data is used - The cloud captures the whole joint distribution: spread, fat tails, skew and how the S&P 500 and VIX move together. A single volatility number captures only the first of these

#### 4.3.2 Step 2

- measure how different the two months are - The wasserstein distnace W2 mathces each point of one cloud to one point of the other so that the total squared distance is as small as possible (hungarian algorithm). $W_2^2 = \frac{1}{21} \min_\pi \sum_i \|X_i - Y_{\pi(i)}\|^2$. Two calm months are close. A calm and a crisis month are far apart becasue the crisis cloud is spread wider.

#### 4.3.3 Step 3

- On the first trading day of each month, k-means with K=2 runs on all windows that end before that day. Windows slide by 1 day, and the history expands - Each cluster's centre is a medoid: the actual window with the smalles total distance to the other windows in its cluster. - it starts with k-means ++ (10 restarts), then alternates assigning windows to the nearest centre and updating the medoids until the asignments stop changing. - the crisis cluster is the one whose windows have the higher average S&P variance

#### 4.3.4 Step 4

- score each day - on day d, the window ending d-1 is compared with both centres: $p_{\text{crisis}} = W_2(\text{calm})^2/(W_2(\text{calm})^2 + W_2(\text{crisis})^2)$ - example: if the distance to calm is 2 and to crisis is 1, $p_{\text{crisis}} = 4/(4+1) = 0.8$ - $p_{\text{crisis}} > 0.5$ exactly when the window is closer to the crisis centre, which is the paper's nearest centroid rule. So MVO and HRP switch exactly as the paper would classify - it uses data through d-1 only, teh same timing as teh other two detectors.

# 5 Regime Interpretation

## 5.1 MVO

- it reads teh detector's daily csv and takes teh value for rebalance date d, which uses data through d-1 - before the detector's first value, $p_{\text{crisis}}$ is set to 0.5, meaning no information - with `DETECTOR = "none"`, $p_{\text{crisis}}$ is 0, every model is its normal version

## 5.2 MVO

- on each rebalance date d, `in_crisis(d)` checks $p_{\text{crisis}}(d) > 0.5$ - if yes, the weights are minimum-variance, if no, max-sharpe - the choice is held until the next rebalance date. - it doesnt matter if it is 0.51 or 0.99, same effect

## 5.3 HRP

- on each rebalance date d, `in_crisis(d)` checks $p_{\text{crisis}}(d) > 0.5$ - if yes, the weights are minimum-variance, if no, max-sharpe - the choice is held until the next rebalance date. - it doesnt matter if it is 0.51 or 0.99, same effect

## 5.4 XGB and RF

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# 6 Regimes

## 6.1 Changepoint Detection

Methodology and calculation 1:1 to the paper

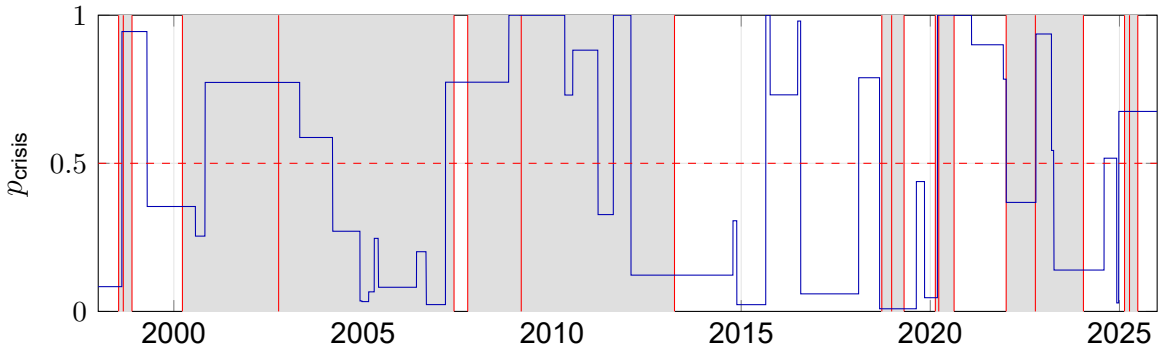



Figure 1: Changepoint detection (VIX): crisis probability $p_{\text{crisis}}$.

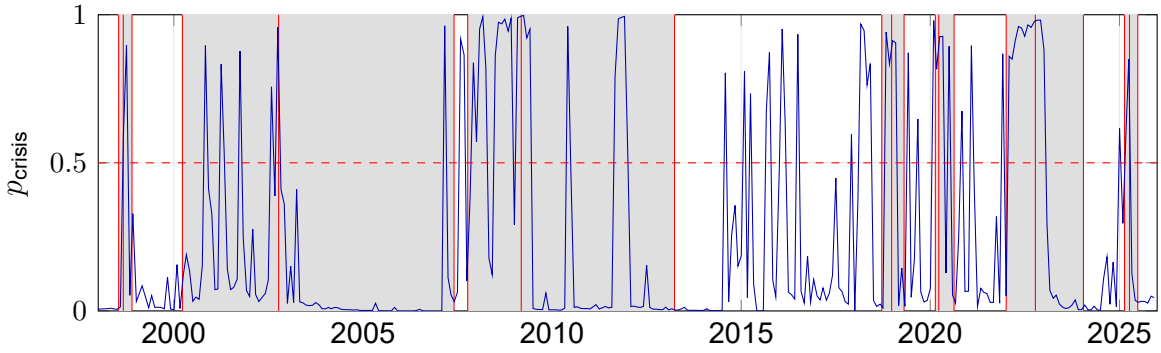



Figure 2: Hidden Markov Model (S&P 500): crisis probability $p_{\text{crisis}}$.

### 6.2 HMM

### 6.3 Wasserstein

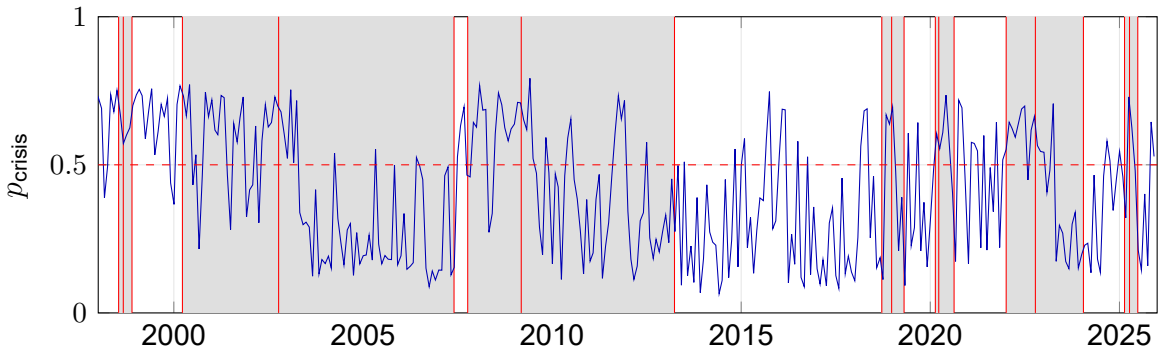



Figure 3: Wasserstein clustering (S&P 500 and VIX): crisis probability $p_{\text{crisis}}$.

### 6.4 Regimes vs. S&P 500

## 7 Portfolio Performance

Table 1: Full-period performance per model, sorted by Sharpe ratio.

| Model | CAGR (%) | Volatility (%) | Sharpe | Max. DD (%) | Turnover p.a. (%) |
|---|---|---|---|---|---|
| MCap | 10.49 | 20.31 | 0.388 | −52.33 | 16.19 |
| Equal Weight | 10.07 | 19.70 | 0.380 | −55.62 | 76.64 |
| MVO | 9.84 | 19.35 | 0.377 | −54.06 | 208.99 |
| HRP | 9.41 | 17.17 | 0.402 | −45.58 | 129.53 |
| XGB | 10.37 | 21.67 | 0.359 | −61.58 | 549.42 |
| RF | 10.51 | 21.95 | 0.360 | −61.00 | 482.94 |
| LSTM | 10.66 | 21.84 | 0.368 | −61.47 | 336.02 |

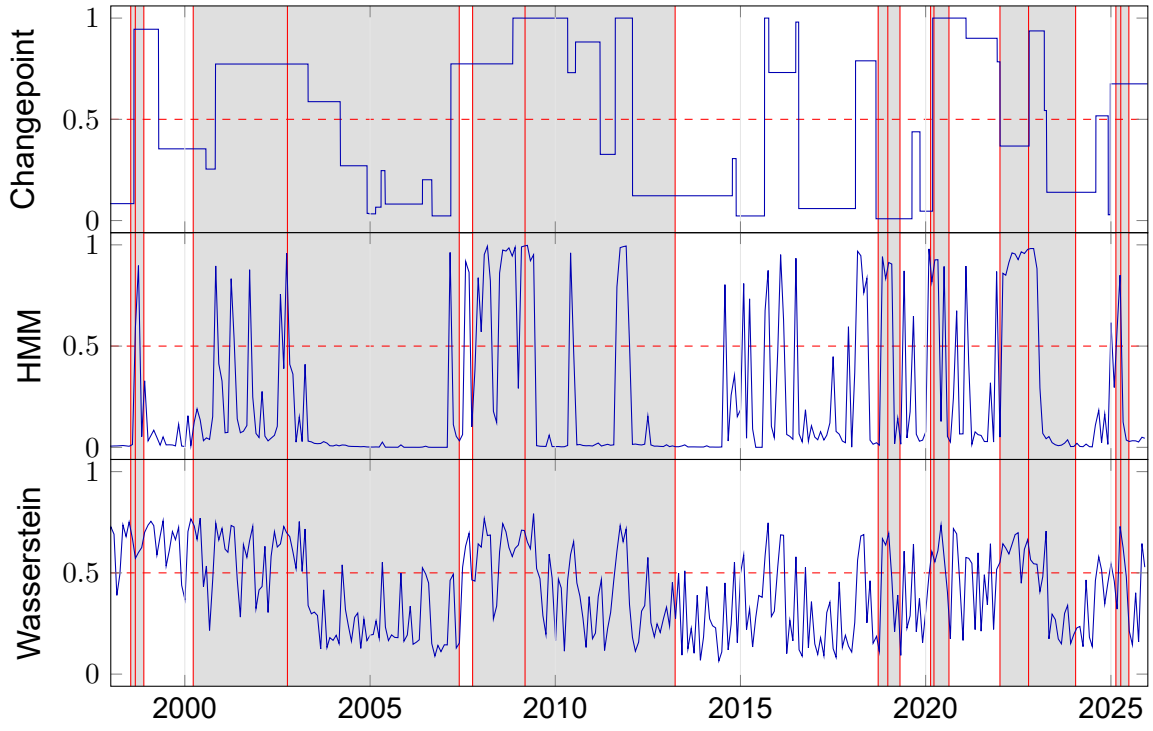



Figure 4: Crisis probability $p_{\text{crisis}}$ of the three regime detectors: changepoint detection (VIX, top), Hidden Markov Model (S&P 500, middle) and Wasserstein clustering (S&P 500 and VIX, bottom). Dashed: 0.5 threshold; grey shading: crisis periods; red lines: peak, trough and recovery.

Table 2: Crisis profile of the baseline models, monthly rebalancing, equal-weighted across seven crises (C1–C7). PTT (peak to trough), TTP (trough to recovery) and Full: average excess return over MCap, $(1+r)/(1+r_{\text{MCap}})-1$. $\Delta$ Ulcer: difference in Ulcer index to MCap, in percentage points (crisis peak to next crisis peak). $\Delta$ Rec.: difference in trading days from the S&P 500 peak until the portfolio regains its starting value (C2 excluded, as MCap did not recover before the GFC; weights rescaled over C1 and C3–C7). Negative $\Delta$ values indicate less pain / faster recovery than MCap, whereas for PTT, TTP and Full positive is better. Hits: crises in which the model's excess return in the phase was positive.

| Model | PTT (%) | TTP (%) | Full (%) | $\Delta$ Ulcer (pp) | $\Delta$ Rec. (days) | Hits PTT | Hits TTP |
|---|---|---|---|---|---|---|---|
| MCap | 0.00 | 0.00 | 0.00 | 0.00 | 0 | – | – |
| Equal Weight | 2.96 | −0.17 | 2.76 | −2.34 | 15 | 4/6 | 3/6 |
| MVO | 1.34 | −0.39 | 0.97 | 0.17 | −82 | 3/6 | 2/6 |
| HRP | 12.41 | −4.26 | 8.07 | −4.70 | −30 | 5/6 | 2/6 |
| XGB | −0.62 | 4.14 | 3.08 | −1.70 | 22 | 3/6 | 3/6 |
| RF | −1.29 | 5.07 | 3.70 | −1.34 | 33 | 3/6 | 4/6 |
| LSTM | −0.85 | 5.39 | 4.00 | −1.40 | 31 | 3/6 | 3/6 |

Table 3: Return (Ret.) and maximum drawdown (MDD) per crisis, in %, monthly rebalancing, from the S&P 500 peak until the S&P 500 is back at that peak: C1 1998 (Russia/LTCM), C2 dot-com, C3 GFC, C4 monetary policy shock 2018, C5 COVID-19, C6 inflation and rate hikes 2022, C7 trade policy shock 2025. GSPC: S&P 500 price index, its return is about zero by construction of the windows.

| | C1 | | C2 | | C3 | | C4 | | C5 | | C6 | | C7 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Ret. | MDD | Ret. | MDD | Ret. | MDD | Ret. | MDD | Ret. | MDD | Ret. | MDD | Ret. | MDD |
| GSPC | 0.1 | −19.3 | 0.2 | −49.1 | 0.3 | −56.8 | 0.1 | −19.8 | 0.1 | −33.9 | 0.9 | −25.4 | 0.5 | −18.9 |
| MCap | 5.6 | −17.0 | −9.1 | −48.6 | 5.4 | −50.4 | 1.9 | −20.2 | 11.1 | −29.0 | 6.1 | −32.4 | 0.5 | −22.4 |
| Equal Weight | 6.9 | −17.0 | 14.2 | −39.3 | 2.3 | −55.6 | 2.6 | −16.7 | 0.9 | −30.8 | 9.0 | −29.1 | −0.8 | −18.7 |
| MVO | 4.5 | −17.5 | −14.5 | −54.1 | 25.6 | −44.6 | −0.7 | −19.9 | 4.8 | −31.6 | 9.0 | −28.7 | −2.3 | −19.2 |
| HRP | 7.5 | −15.6 | 31.1 | −34.3 | 25.5 | −45.6 | 2.0 | −14.5 | 0.1 | −29.6 | 5.6 | −23.8 | −4.5 | −15.1 |
| XGB | 3.5 | −19.6 | 17.0 | −38.6 | −3.1 | −61.6 | 2.8 | −18.3 | −0.5 | −32.3 | 12.7 | −32.9 | −1.0 | −18.9 |
| RF | 3.4 | −19.9 | 20.7 | −39.6 | −5.8 | −61.0 | 5.1 | −17.8 | −4.0 | −34.1 | 13.9 | −33.8 | 1.3 | −19.7 |
| LSTM | 6.8 | −17.8 | 23.7 | −42.2 | −7.9 | −61.5 | 5.3 | −16.3 | −1.4 | −32.1 | 17.1 | −33.8 | −0.6 | −17.1 |

## 7.1 Crisis Excess Returns: Peak to Trough vs. Trough to Recovery

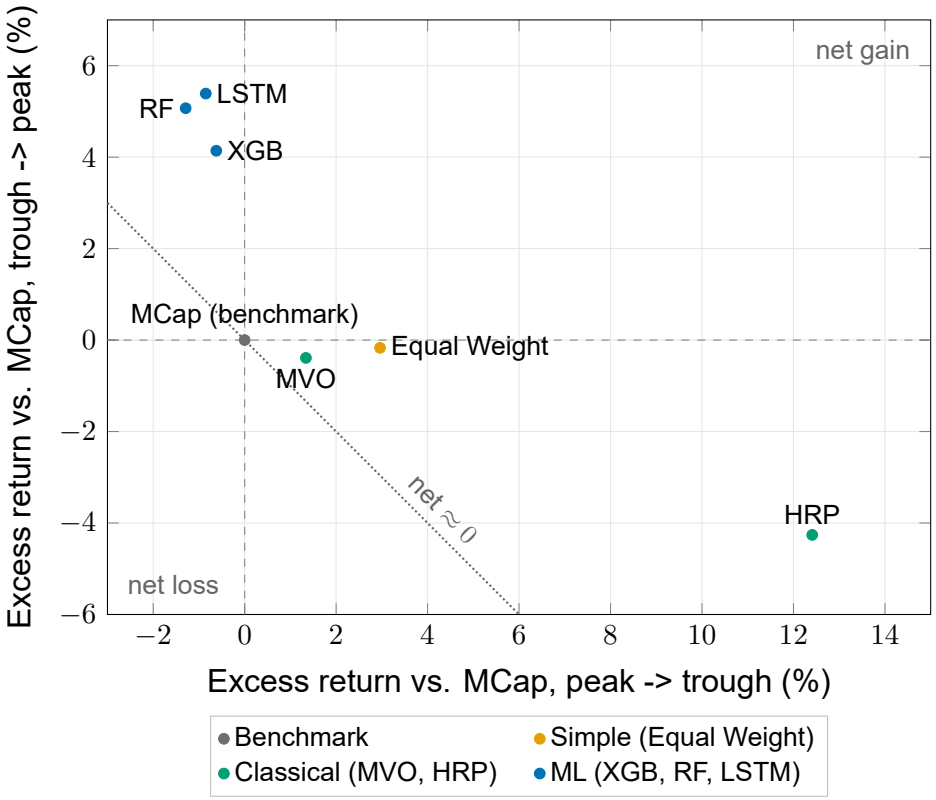



Figure 5: Average crisis excess return over the benchmark per model: peak to trough ($x$) vs. trough to recovery ($y$). The dotted line marks $x + y = 0$: models above it gain over the whole crisis (peak to recovery), models below lose. The line is approximate, since the two phases compound, $(1 + x)(1 + y) - 1 \neq x + y$, rather than add.

## 7.2 Detection Quality

Table 4: Crisis detection quality of the regime detectors by evaluation frequency. Defensive: share of days in the crisis state; Episodes: number of crisis-state episodes; Avg. ep.: mean length of an episode in trading days; Recall: share of crisis days flagged; Precision: share of flagged days that are crisis days; Lift: precision relative to the share of crisis days in the sample; False alarms: flagged days outside crises; Missed: crises (of seven) never flagged.

| Detector | Defensive (%) | Episodes | Avg. ep. (days) | Recall (%) | Precision (%) | Lift | False alarms (days) | Missed |
|---|---|---|---|---|---|---|---|---|
| *Daily* | | | | | | | | |
| Changepoint | 48.72 | 10 | 343 | 67.45 | 25.73 | 1.38 | 399 | 1 |
| HMM | 23.24 | 156 | 10 | 57.44 | 45.93 | 2.46 | 397 | 0 |
| Wasserstein | 41.08 | 110 | 26 | 75.09 | 33.97 | 1.82 | 970 | 0 |
| *Monthly* | | | | | | | | |
| Changepoint | 48.73 | 9 | 381 | 68.14 | 25.99 | 1.39 | 397 | 2 |
| HMM | 22.26 | 32 | 49 | 47.97 | 40.05 | 2.15 | 308 | 0 |
| Wasserstein | 39.71 | 41 | 68 | 72.88 | 34.10 | 1.83 | 957 | 0 |
| *Quarterly* | | | | | | | | |
| Changepoint | 49.09 | 9 | 384 | 68.67 | 26.00 | 1.39 | 441 | 1 |
| HMM | 21.47 | 15 | 101 | 42.24 | 36.57 | 1.96 | 255 | 2 |
| Wasserstein | 42.89 | 21 | 144 | 64.78 | 28.07 | 1.50 | 440 | 2 |
| *Yearly* | | | | | | | | |
| Changepoint | 53.50 | 6 | 628 | 73.94 | 25.69 | 1.38 | 252 | 2 |
| HMM | 17.79 | 5 | 251 | 21.92 | 22.90 | 1.23 | 0 | 4 |
| Wasserstein | 39.22 | 8 | 345 | 59.12 | 28.02 | 1.50 | 252 | 2 |

Table 5: Timing of the regime detectors around the crises, by evaluation frequency. Exit lag: median trading days from the crisis peak until the detector switches to the crisis state (negative = before the peak); Re-entry lag: median trading days until it switches back; Missed: crises not flagged; Late def.: crises flagged only late; Not recovered: crises after which the detector did not leave the crisis state before the next crisis.

| Detector | Exit lag (days) | Re-entry lag (days) | Missed (ptt) | Late (ttp) | Still defensive |
|---|---|---|---|---|---|
| *Daily* | | | | | |
| Changepoint | $-34$ | $406$ | C3 | – | C4, C6 |
| HMM | $7$ | $24$ | – | – | – |
| Wasserstein | $3$ | $36.5$ | – | – | – |
| *Monthly* | | | | | |
| Changepoint | $-122.5$ | $373$ | C3, C4 | C4 | C4, C6 |
| HMM | $19.5$ | $47.5$ | – | – | – |
| Wasserstein | $19.5$ | $85$ | – | – | – |
| *Quarterly* | | | | | |
| Changepoint | $-120$ | $158$ | C4 | C4 | C4, C6 |
| HMM | $61$ | $58.5$ | C3, C4 | C3, C4 | – |
| Wasserstein | $-27.5$ | $163$ | C3, C4 | C3, C4 | – |
| *Yearly* | | | | | |
| Changepoint | $14.5$ | $563$ | C3, C4 | – | C6 |
| HMM | $14.5$ | $-43$ | C1, C3, C4, C5 | C3, C5 | C6 |
| Wasserstein | $30$ | $209$ | C3, C4 | C3 | C6 |

## 7.3 LSTM (No Regime)

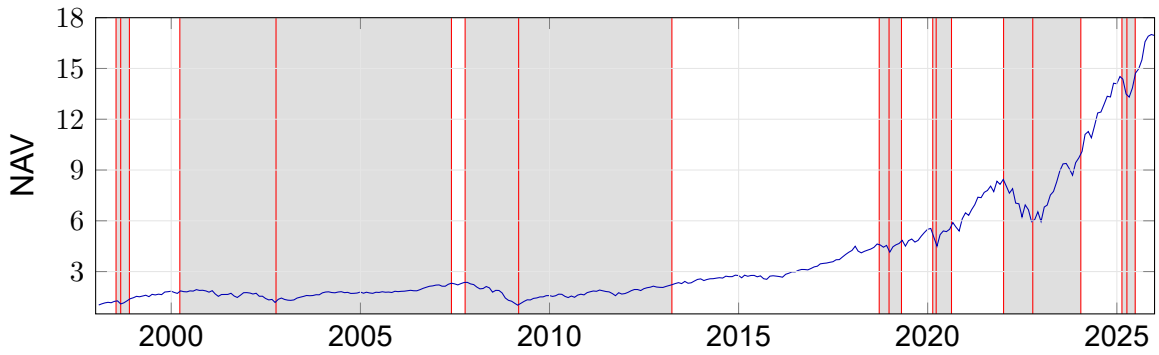



Figure 6: LSTM ($t = 10$, holdout, rank) without regime overlay: monthly portfolio value (NAV).

## 7.4 LSTM (Changepoint VIX)

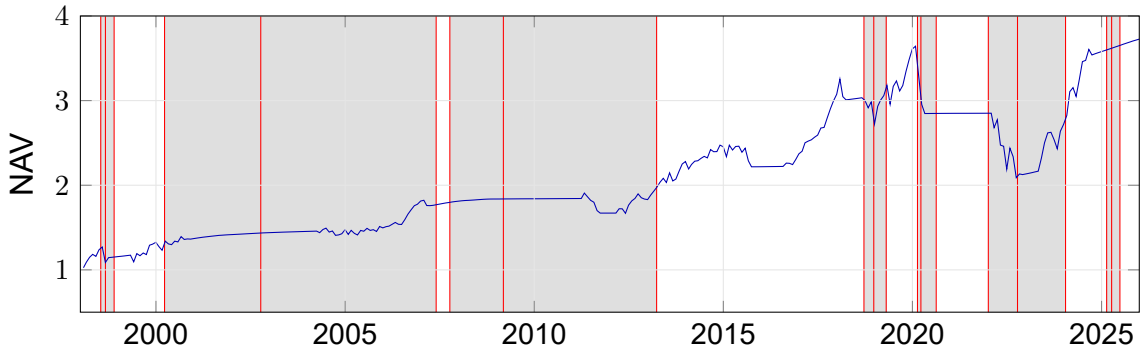



Figure 7: LSTM ($t = 10$, holdout, rank) with changepoint (VIX) regimes and cash: monthly portfolio value (NAV).

## 7.5 LSTM (Changepoint VIX, Cash)

## 7.6 LSTM (HMM S&P 500)

## 7.7 LSTM (Wasserstein S&P 500 and VIX)

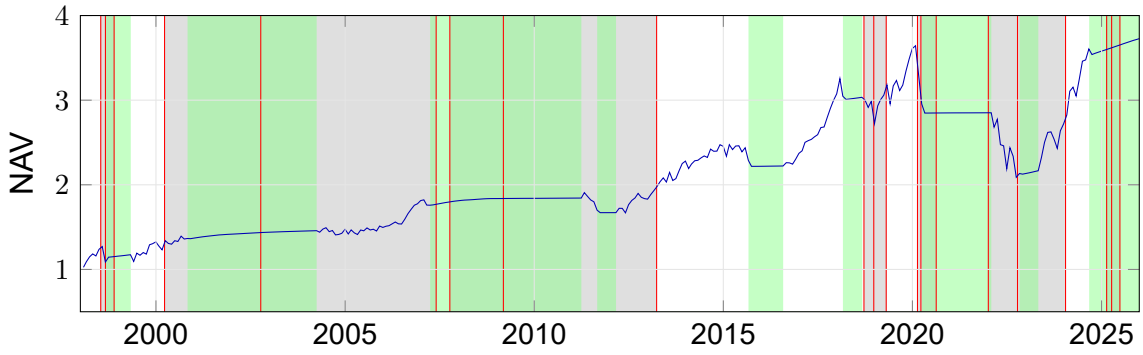



Figure 8: LSTM ($t = 10$, holdout, rank) with changepoint (VIX) regimes and cash: monthly portfolio value (NAV). Green: months held in cash.

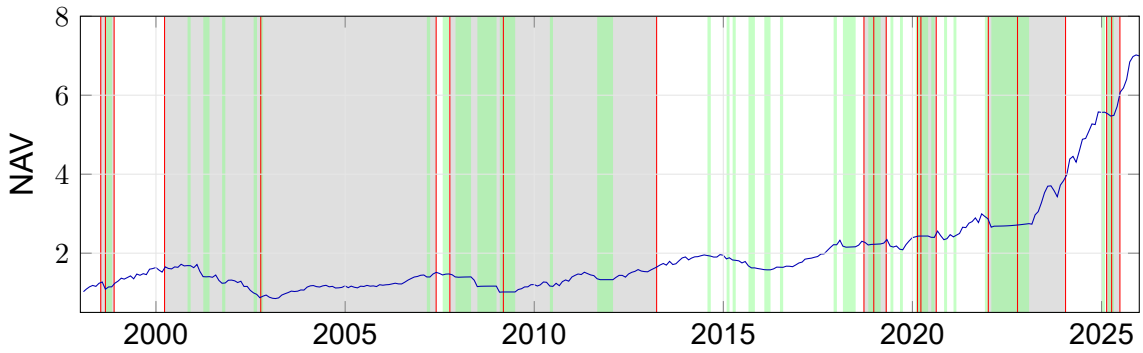



Figure 9: LSTM ($t = 10$, holdout, rank) with HMM (S&P 500) regimes and cash: monthly portfolio value (NAV). Green: months held in cash.

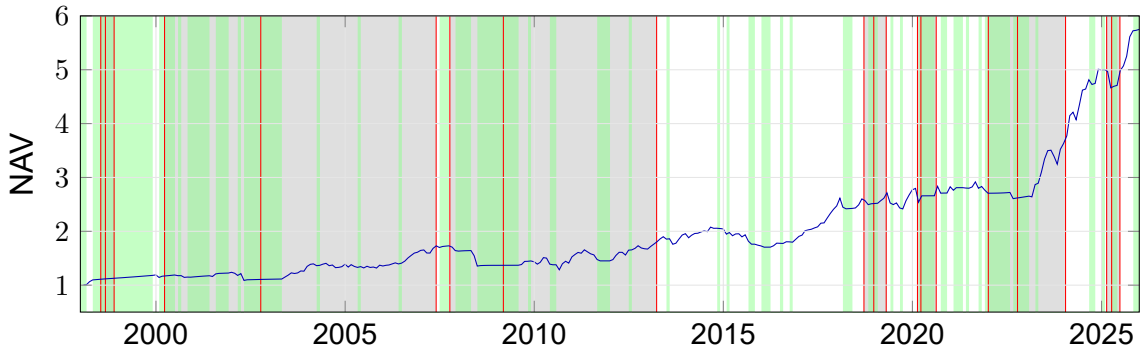



Figure 10: LSTM ($t = 10$, holdout, rank) with Wasserstein (S&P 500 and VIX) regimes and cash: monthly portfolio value (NAV). Green: months held in cash.

# 8 References

Nystup Peter, Madsen Hendrik, Lindström Erik, 2017. Dynamic portfolio optimization across hidden market regimes.
Used for:
- Transaction costs 10bps
- Hidden Markov Model